THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 9 — SCIENCE · HARD LEVEL CHAPTER TEST SET 1

Chapter 2: **Exploring Mixtures and Their Separation** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Assertion (A): Milk shows the Tyndall effect but salt solution does not. Reason (R): The particles of milk are larger than the particles of a true solution. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

2. A solution has 30 g of solute in 120 g of solvent. 90 g of the solvent is evaporated. The new percentage by mass of solute is: 1

(a) 75 per cent **(b)** 50 per cent **(c)** 20 per cent **(d)** 25 per cent

3. A mixture can be separated by sublimation if: 1

(a) only one component sublimes **(b)** both sublime at equal rates
(c) neither component melts **(d)** both melt

4. Assertion (A): Chromatography can separate the colours of a black ink. Reason (R): Different dyes have different solubilities in the solvent. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

5. The solubility of KNO3 is 64 g/100 g water at 40 degrees Celsius and 32 g at 20 degrees Celsius. 164 g of saturated solution at 40 degrees Celsius is cooled to 20 degrees Celsius. The mass of crystals is: 1

(a) 16 g **(b)** 32 g **(c)** 96 g **(d)** 64 g

6. Which of the following combinations can be separated by magnetic separation? 1

(a) water and oil **(b)** sand and chalk **(c)** iron and sulphur **(d)** salt and sugar

7. Assertion (A): Fractional distillation is used to separate the components of air. Reason (R): The gases in liquid air have different boiling points. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

8. A mixture contains 30 g of NaCl and 50 g of KNO3. Both dissolve in 100 g of hot water at 80 degrees Celsius. On cooling to 20 degrees Celsius (solubility NaCl 36, KNO3 32), the mass of KNO3 crystallising out is about: 1

(a) 32 g **(b)** 14 g **(c)** 18 g **(d)** 50 g

9. **Assertion (A):** A mixture of salt and sand can be separated by dissolving it in water, filtering and evaporating the filtrate. 1
Reason (R): Sand is insoluble in water, whereas salt is soluble in water.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. **Assertion (A):** Fractional distillation is used to separate two miscible liquids whose boiling points differ by less than 25 K. 1
Reason (R): Simple distillation can separate a volatile liquid from a non-volatile dissolved solid.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. A solution has density 1.2 g/mL and contains 24 g of solute in 200 mL. Find the mass by mass percentage. 2

12. Explain why the thermometer reading stays constant during distillation of pure water. 2

13. Why is the sugar crystal formed by slow cooling larger than that formed by rapid cooling? 2

14. Why is it possible for a solution to be both dilute and saturated? Give an example. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. The solubility of a salt is 25 g per 100 g water at 20 degrees Celsius and 60 g per 100 g water at 60 degrees Celsius. Find the mass of crystals obtained when 320 g of saturated solution at 60 degrees Celsius is cooled to 20 degrees Celsius. 3

16. A student prepared two solutions: 5 g sugar in 50 g water (A) and 8 g sugar in 100 g water (B). Compare the concentrations. 3

17. Explain the 'salting out' of a colloidal sol such as soap using sodium chloride. 3

OR

In a chromatogram, three components X, Y, Z travel 2.0 cm, 4.5 cm and 6.0 cm while the solvent travels 7.5 cm. Find their R_f values and name the most strongly adsorbed. 3

18. Explain how forensic scientists use chromatography to identify ink used in a forged document. 3

OR

Explain how one can test whether a colourless liquid is pure water. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** A student in Pune dissolves 20 g of common salt in 180 g of water to prepare a solution for an experiment. 5
(i) Name the solute and the solvent.
(ii) Find the mass of the solution.
(iii) Find the mass percentage of salt in the solution.
(iv) How much more water must be added to this solution to make a 5 per cent (by mass) solution?

20. Explain how the principle of chromatography, the concept of R_f and the effect of the solvent choice may be used to identify unknown dyes in food colours. 5

OR

A town gets contaminated river water containing mud, bacteria and dissolved salts. Plan a treatment scheme with justification of each method and its principle. 5

END OF QUESTION PAPER — Total: 40 Marks